GNSS Positioning Under Threat: The Rising Risk to Existing Systems and The Role of Alternative Indoor and Seamless Navigation Technologies

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***Abstract*—What began with isolated incidents of GPS interference has grown into a global crisis that threatens everything from commercial aviation to military operations. This article documents the alarming reality and the importance of Global Navigation Satellite System (GNSS)-denied navigation technologies in this context. GNSS attacks have increased sevenfold in contested regions, rendering precision-guided weapons nearly useless and forcing airlines to abandon entire routes. The availability of some inexpensive jammers (less than $50) has meant that the devices used by countries can be easily acquired by anyone, significantly compromising security. We present an investigation of this escalating threat through numerous comprehensive real-world case studies, including air traffic chaos over the Baltic Sea, maritime spoofing in international waters and the failure of precision weapons in active conflict zones. In addition, we share an 819-minute open-source dataset of experimental GNSS raw data, featuring three different types of attacks on a GNSS receiver under various motion conditions. The analysis of the main impact of these attacks on the raw measurements at the receiver level and a summary of the footprint of each attack based on the measurements is also provided. Finally we explain how the positioning and navigation solutions developed for indoors offer decisive advantages for mitigating these attacks, solving outdoor navigation vulnerabilities. This research shows that the future of secure navigation lies not in hardening satellite systems, but in making them optional.**

***Index Terms*—Alternative indoor positioning methods, global navigation satellite system (GNSS) interference, jamming, spoofing.**

I. Introduction

RADIO interference techniques such as jamming, spoofing, and meaconing [1], originally developed in military conflicts, now significantly impact modern navigation technologies. Dating back to World War I, when the British and French jammed the navigation of German Zeppelins and further refined during World War II, when the Royal Air Force manipulated Luftwaffe bomber systems, these methods have evolved into sophisticated tools [2]. Originally developed to gain strategic advantage in warfare, they now represent serious vulnerabilities in civilian applications.

They have expanded in their application and impact, as recent conflicts, such as the Russian invasion of Ukraine and the conflicts between Israel and Hamas and Hezbollah, demonstrate [3]. Both state and nonstate actors use spoofing and jamming to disrupt communications, interfere with surveillance and manipulate the reliability of technologies that depend on Global Navigation Satellite System (GNSS). They affect the accuracy of GNSS-based weapons such as surveillance drones or deny reliable use for basic navigation in urban areas. For example, while the Ukrainians hit their target 50% of the time at the beginning of the war with 155 mm Excalibur shells whose Global Positioning System (GPS) coordinates were programmed before firing, this rate fell to less than 10% in the following months due to GPS jamming [4].

The impact of such interference extends beyond military operations to affect civilian GNSS usage as well. A notable example occurred on April 19, 2017, when a GPS jammer left plggued into the cigarette light of a car at Nantes airport disrupted the navigation systems of nearby airplanes, causing delays of up to 1 h and 15 minutes before takeoff. The individual responsible was fined €2000 by the Nantes Criminal Court for causing significant disruption [5].

Given the increasing sophistication and frequency of GNSS jamming, meaconing, and spoofing, truly resilient Positioning, Navigation, and Timing (PNT) solutions have become an operational imperative, especially in dense urban canyons, critical infrastructure zones and indoor settings where satellite signals are weakest. These vulnerabilities have triggered a wholesale reexamination of alternative technologies that, although originally refined for indoor use, now offer decisive advantages outdoors as well. Tightly coupled sensor-fusion architectures, advanced signal processing techniques, and robust integrity

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monitoring algorithms are no longer optional add-ons; they form the backbone of next-generation PNT systems that can resist hostile RF conditions.

This study is an extension of a conference proceeding [6] recalling the increasing attacks on GNSS from 2017 to 2025 with new examples of their impact in different application areas (Aviation, Maritime, Automotive and Military). The study draws on incident reports from reputable news agencies, official aviation and maritime reports and peer-reviewed literature. We have added an new analysis (Section IV) of GNSS attacks using a new experimental dataset that covers all three types of GNSS attacks under different dynamic conditions. We also publish this 819-min dataset in a context where the acquisition of such data is technically difficult and legally restricted in many countries. Both the raw measurements and the analysis code are made available as open source for further research. The analysis distinguishes the footprint for each attack, how it is affected by the attacks and how we can recognise differences between them. Finally, the article reviews recent advances in indoor positioning technologies developed within the Indoor Positioning and Indoor Navigation (IPIN) community and highlights their utility for addressing outdoor navigation vulnerabilities and providing robust PNT capabilities.

II. Taxonomy of GNSS Attacks on Positioning and Navigation

GNSS-based positioning and navigation attacks encompass a range of tactics used to disrupt or deceive communications and navigation systems, with jamming, spoofing, and meaconing being the three most important methods. They are used in both the military and civilian sectors and pose a major challenge to the functioning and integrity of modern technical infrastructure and navigation systems.

A. Three Main Types of Attacks

1) Jamming: GNSS jamming involves transmitting radio noise on the same frequency as the GNSS signals to overpower the receiver with useless data and make the real signal indistinguishable. This approach can affect various systems, including signal of opportunity, radar, GNSS, and other satellite-based communication channels.

There are different types of jamming. Spot jamming focuses on a specific frequency, while barrage jamming covers a broad spectrum of frequencies simultaneously. The systems that interfere with communication terminals and GNSS receivers in a local area are downlink jammers, while those that interfere with the satellites themselves and thus affect services for all users in the satellite reception area are uplink jammers.

Efficient jamming involves maximizing overlap with the desired signal subspace while avoiding wasteful power allocation both outside and directly in the passband of the target receiver to extend battery life. Interfering signals that are continuous in the time domain and white in the frequency domain within the passband of the signal are reasonably efficient as they largely overlap the desired signal subspace. For interference signals with a matched spectrum, the spectrum of the interference signal can be shaped to match that of the desired GNSS signal, making it difficult to filter out the interference without also removing the desired signals.

2) Spoofing: Spoofing involves broadcasting counterfeit GNSS signals to deceive the receiver about its position and/or time. Compared to jamming, spoofing signals match the code interference with all the additional modulations required to make the interference signal's structure and content identical to an authentic signal. As a result, false data is generated that appears legitimate to the receiver leading to significant disruptions: Misguiding ships, drones, or other vehicles. The forms of spoofing can be manifold. How best to protect against spoofing interference remains an open problem.

3) Meaconing: Meaconing is the rebroadcast of authentic GNSS signals either previously recorded or relayed from another location, which may be delayed or otherwise manipulated. Although it can be considered a technological subset of spoofing, it leaves a distinctive signature on the receiver, as we see in our practical analysis and is treated as a standalone attack's type in the literature [1].

B. Vulnerability Points of GNSS Signals

The primary vulnerability of GNSS signals is their low power. GNSS signals are broadcast in L-band ($\sim$1.1–1.6 GHz) from satellites in medium Earth orbit, which typically results in a $[-130, -125]$ dBm on Earth, compared with $[-85, -50]$ dBm for cellular signals. This makes GNSS receivers particularly susceptible to being easily overpowered by local sources of interference. Attack technology typically involves signal generators and amplifiers that can generate signals at the same frequencies but with higher power. The second vulnerability lies in their lack of encryption, making them an easy target for spoofing. Finally, it has been shown that GNSS receivers are more vulnerable to jamming and spoofing during cold start, as they lack the necessary ephemeris to distinguish between authentic and interference signals.

III. Case Study Analysis

A. Report on Escalating GNSS Attacks

The trend towards the use of jammers and the sharp increase in GNSS attacks continues. As early as 2019, 450 000 GNSS jamming signals were recorded as part of the EU-funded STRIKE3 project [7], which has set up a worldwide GNSS interference monitoring network in 23 countries. 73.000 were classified as "significant impact on GNSS" and 59 000 were identified as "jamming signals". The analysis shows that more than 10% of all interference could be malicious or intentional.

Low Earth Orbit (LEO) satellites can also be used to detect jammers. From 2017 to 2020, the software-defined GNSS receiver FOTON on board the International Space Station (ISS) monitored GNSS jamming and observed the increasing frequency and sophistication of GNSS attacks [8]. The innovative method captured raw signal data and used Doppler positioning to accurately locate the sources of interference. A key finding was the persistent L1&L2 interference from a source in Syria, which has been active since 2017, as well as in China. Since late 2019, Spire Global's CubeSat constellations have also been used to monitor GPS L1&L2 jamming by analyzing the Signal to Noise Ratio (SNR) from precise orbit determination with background noise suppression and radio occultation measurements [9]. Observations indicate that GNSS services in Europe, the Middle East and North Africa are significantly limited due to increasing interference, especially during geopolitical conflicts.

The trend is also increasing because it has become easier to jam. Non-military jammers, which cost no more than 50 US dollars, can already block weak signals from satellites. These devices are assembled using off the shelf equipment available online.

B. Aviation

Two Finnair planes that took off from Helsinki on April 29, 2024, had to turn back because they could not land at Tartu Airport in Estonia due to Russian GNSS interference [10]. Unfortunately, the Estonian city is not yet equipped with facilities enabling aircraft to land without GNSS. As a result, the Finnish airline has decided to suspend its flights to Tartu in Estonia for a month. This incident joins a series of intense jamming zones observed in the Baltic Sea, the Black Sea and the Eastern Mediterranean. In recent months, a threshold of 350 perturbations per week has been observed, compared to an average of around 50 in 2023. Between August 2023 and March 2024, a total of 46 000 flights with GNSS navigation disturbances were reported over the Baltic Sea region. Of these, 2309 Ryanair flights, 1368 Wizz Air flights, 82 British Airways flights, and 4 Easyjet flights were affected. In March 2024, GNSS jamming over Lebanon impacted civil aviation severely, forcing a Turkish Airlines flight to return to Turkey instead of landing in Beirut due to GNSS interference.

Circle spoofing was first documented in Ukraine, where it disrupted air-traffic navigation by convincing aircraft that they were at incorrect geographic coordinates. As reviewed in [11], 70 papers on GNSS spoofing in flying ad-hoc networks (FANETs)–co-operative swarms of UAVs–and classifies them into four themes: 1) Attack Mechanisms; 2) Defense Mechanisms; 3) Spoofing Used as a Defense Strategy; and 4) Impact-and-Vulnerability Assessment.

In these attacks, counterfeit GNSS signals are transmitted with sufficiently high power to create a denial-of-service environment. The three primary spoofing strategies are misleading the target to a false destination, misreporting positional information and using techniques such as position denial and track break to fool the drone's navigation systems. Spoofers can operate from various locations, including remote, nearby and onboard positions. They use systems such as simulators, repeaters, and hardware injection to generate spoofed signals. Various strategies are proposed to mitigate GNSS spoofing, including cryptographic methods, blockchain technology and distributed radar systems. However, challenges remain in terms of practicality, cost, and computational complexity.

C. Maritime

Vessels exchange information via the Automatic Identification System (AIS) on the VHF frequencies 161.25 and 162.025 MHz, including the name of the vessel, its GNSS position and its speed/course over ground. Cases where AIS signals have been hijacked or cut off can be seen on websites such as Marine Traffic or Vessel Finder, which track vessels in real time. Disabling AIS signals is motivated by illegal fishing. In Venezuela and Ukraine, vessels used false GNSS coordinates and spoofed the AIS system to circumvent international law and sanctions [12].

In 2013, a maritime spoofing attack experiment was conducted to secretly manipulate GNSS signals and divert the $80 million yacht White Rose of Drachs from its original course [13]. Using a custom-built GNSS spoofing device, the researchers sent out false GNSS signals that gradually overpowered the authentic signals. This allowed them to alter the yacht's course in international waters near Italy, demonstrating how easily a ship's navigation system can be manipulated by spoofing.

During the 2025 Iran–Israel conflict, the persistent GNSS spoofing and jamming in the Strait of Hormuz have repeatedly forced tankers' reported positions to jump onto land or hundreds of nautical miles off their true track. On 17 June 2025, two vessels–the VLCC Front Eagle and the Suezmax Adalynn–collided and caught fire south-east of Fujairah amid a surge of electronic interference in the waterway [14].

D. Automotive

In [15], researchers spoofed the navigation system of an autonomous vehicle, especially when turning. Real GNSS data from RINEX files including pseudoranges were combined with a predetermined spoofed location and processed by a spoofed pseudoemulator. It uses machine learning models or experimental correlations to estimate pseudoranges that match the spoofed location. These pseudoranges are used together with valid clock, ephemeris and integrity variables in a GNSS positioning algorithm to determine the spoofed location. If successful, these pseudoranges are sent to a spoofing GNSS signal simulator, which alters the satellite signals to direct the vehicle's GNSS receiver to the spoofed location. Spoofing signals are gradually combined with the legitimate signals so that they are not detected by the on-board system. This method was tested on a vehicle equipped with a NovAtel CPT7700 and TerraStar corrections. Navigation errors increased under spoofing attacks, with the vehicle turning in the wrong direction.

The authors in [16] focused on the detection and identification of vehicles that emit GNSS jamming signals on roads. They exploit crowdsourced GNSS data from the mobile phones of a group of vehicles passing in front of roadside monitoring stations. The underlying hypothesis is that this group of vehicles should exhibit a similar pattern at successive monitoring stations, allowing the identification of interfering signals. To distinguish fading from interference and natural scenarios (tall buildings, underpasses, tunnels), Carrier-to-Noise Ratio (C/NO) profiles are created at each station. Road experiments with monitoring stations, consisting of a universal software radio

Fig. 1. Visible part of the satellite communication suppression system Tobol.

peripheral (USRP) N210 and a WBX daughterboard with GNU-RADIO scripts, were conducted over several days to investigate how likely it is that a monitoring station can unambiguously identify a jamming vehicle. It was found that in most cases a unique identification is possible with two appropriately spaced monitoring stations. The vehicle is identified via electronic toll systems or camera images on which license plates can be recognized. The experiments demonstrate the risk of jamming in road traffic as well as a robust system for real-time monitoring and identification of GNSS signal tampering.

E. Military

Leaked U.S. intelligence documents reveal significant threats posed by Russia and China to U.S. and allied space assets [17]. These threats encompass advanced methods for jamming communications, disrupting GNSS signals, and direct satellite attacks.

Russia has developed and deployed several electronic warfare systems (R-330Zh Zhitel, Bylina-MM, Krasukha-4) capable of jamming communications satellites and disrupting GNSS signals within a significant range. The Tobol interception device (Fig. 1), manufactured by the Russian company NPP inprokom, consists of a trans-horizon air surveillance radar and a SIGINT jamming antenna system working with two primary strategies [18]. The simplest strategy jams enemy satellite communications. Ground-based or third-party systems identify the target satellites and transmit the target designation to the Tobol complex. Once this information is received, interference is generated to prevent the target satellite from receiving signals from Earth. The second strategy interferes with the signals broadcasted from the satellites to the ground targets. It receives the satellite signal and transmits corresponding interference that prevents it from receiving the intended data from orbit. Deployment in fixed structures enhances the performance characteristics so that even a limited number of units can effectively monitor and protect the country's borders and surrounding areas. According to [19], there were two exceptional incidents caused by the Tobol around Kaliningrad on January 10 and 16, 2024. A good half of Poland as well as some areas in Sweden and Lithuania were affected by this interference on both occasions. The system also managed to desynchronize Starlink's GNSS-based communication. Software updates were implemented to counteract these attacks.

Recently, a Russian aircraft, which was engaged in bombing operations with guided bombs over Kharkiv, dropped its bomb on the Russian city of Belgorod, landing in a residential area, causing significant damage, probably due to a GNSS attack [20]. The consequences of this accidental hit were considerable: 30 private houses and 10 vehicles were damaged. This incident highlights the risks and unpredictability of military operations in populated areas.

China has significantly expanded its space capabilities and doubled the number of its satellites since the founding of the U.S. Space Force. It has over 700 operational satellites, of which about 250 are used for intelligence, surveillance and reconnaissance (ISR). In the event of a conflict, for example over Taiwan, they could support jamming GNSS, communications and reconnaissance satellites, interfere with space ground networks, and impact ballistic missile early warning satellites.

These examples highlight the growing threat of GNSS jamming and spoofing across all sectors and emphasize the need for robust countermeasures and alternative navigation aids to ensure safety and operational integrity.

IV. Real World GNSS Attack Experiments:

To understand the impact of these attacks on a GNSS receiver, it is interesting to analyse what happens to the experimental data during the attack. This analysis, based on the data collection carried out during the Jammertest campaign in autumn 2024 [21], can support the development of methods to detect and mitigate attacks (Section V).

A. Experimental Scenarios

The data presented in this article was collected in northern Norway and includes different types of attacks (Jamming, Meaconing and Spoofing) carried out in different scenarios with dynamic and stationary rovers. The total attack time for this dataset is 819 minutes. Different power levels were used in the experiments and different frequency bands and GNSS constellations were attacked. Our dataset is detailed as follows.

1) **Jamming**—7 scenarios (6 dynamic, 1 stationary): 430 min dynamic /14 min stationary (444 min total)
2) **Spoofing**—8 scenarios (3 dynamic, 5 stationary): 64 min dynamic /155.5 min stationary (219.5 min total)
3) **Meaconing** – 5 scenarios (2 dynamic, 3 stationary): 28 min dynamic /46 min stationary (74 min total)
4) **Spoofing + Jamming**—4 scenarios (2 dynamic, 2 stationary): 40.5 min dynamic /41 min stationary (81.5 min total)

B. Attack Generation Hardware Setup

The experimental setup incorporated a variety of jamming devices to assess the resilience of GNSS receivers under different attack conditions. Table I summarizes the jammers used, ranging from low power handheld units (H6.1, H6.2, H6.3, and H6.5) that selectively disrupt specific L-band channels, to a high-power jammer (F8.1) capable of simultaneous multiband interference with up to 50 Watts Equivalent Isotropic Radiated Power (EIRP).

Regarding spoofing, a Software Defined Radio (SDR) (BladeRF x115 from Nuand) was used for mobile spoofing tests.

TABLE I
SUMMARY OF JAMMER HARDWARE SPECIFICATIONS

Model	Freq (MHz/BW)	Bands	Output	Sweep (μs)
H6.1	1580 (20)	L1, E1, B1C	28 dBm	6
H6.2	1580 (28), 1155 (105), 1248 (109)	L1, E1, B1C; L5, G3, B2a/b, E5a/b; L2, G2, G3, B2b, B3l, E5b, E6	30/26/28 dBm	7
H6.3	1580 (25), 1152 (108), 1246 (107)	L1, E1, B1C; L5, G3, B2a/b, E5a/b; L2, G2, G3, B2b, B3l, E5b, E6	30/26/28 dBm	7
H6.5	1176 (8.6), 1247 (80), 1593 (80)	L5, B2a, E5a; L2, G2, B3l, E6; L1, E1, B1C, B1l, G1	30/32/31 dBm	10
F8.1/Porcus Major	Multi-channel	8 GNSS bands	Up to 50 W EIRP	N/A

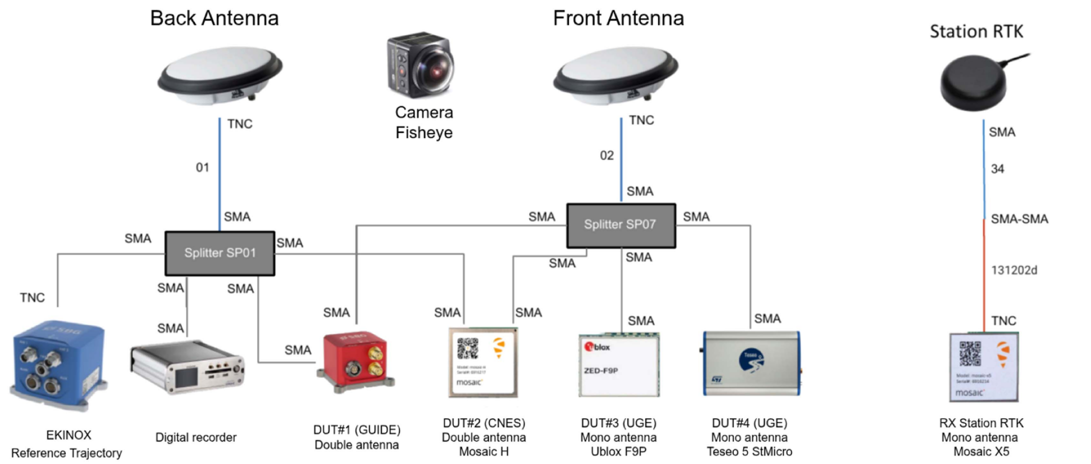


Fig. 2. Roof-mounted GNSS setup: two dual-frequency antennas split to feed the reference INS, digitizer, four DUT receivers (GUIDE, Mosaic H, ZED-F9P, Teseo 5), with a separate Mosaic X5 RTK base.

The output signal is amplified by 45 dB through an amplifier, achieving a maximum total EIRP of approximately 10 dBm. This signal is transmitted through a dipole antenna mounted on the roof odf the vehicle (Fig. 6).

The F1.1 system is used for meaconing. The system consists of two GNSS receiving antennas (E1 and E2), which are located at different locations and each have a distance to the transmitting antenna. The real sky signals from these antennas are transmitted via long cables and retransmitted via a directional antenna (E3) aimed at the community house in Bleik (Fig. 5). The receiving antennas are located outside the Line of Sight (LOS) of the transmitter to prevent a feedback loop. This configuration supports switching between the two receiving antennas, adjusting power levels and transmitting both signals simultaneously.

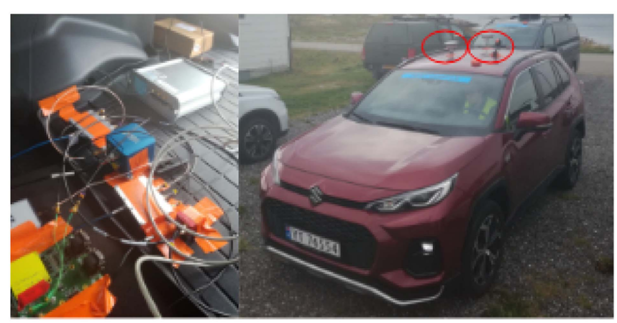


Fig. 3. Test setup for the test vehicle. Left: The internal test equipment, including GNSS receivers and supporting electronics installed inside the vehicle. Right: The exterior view of the test vehicle with the GNSS antennas mounted on the roof (highlighted by red circles).

C. Receiver and Data Logging Setup

Fig. 2 details the GNSS signal distribution to onboard receivers and Fig. 3 presents the external test vehicle setup with roof-mounted instrumentation. Two dual-frequency GNSS patch antennas and a fisheye camera are fixed side-by-side on the roof using vibration-damping brackets and sealed bulkheads. Each antenna terminates in a TNC connector and is cabled via low-loss coaxial lines through a roof bulkhead to RF splitters. The rear antenna is routed through an SMA-to-TNC adapter into splitter SP01, which distributes its output as follows.

1) Ekinox Inertial Navigation System (INS)/GNSS reference unit.
2) High-speed digitizer.
3) GUIDE dual-antenna receiver (DUT #1).

The front antenna feeds the SP07 splitter, which transmits the signals to the following.

1) Mosaic H dual-antenna receiver (DUT #2),
2) U-blox ZED-F9P single-antenna receiver (DUT #3),
3) Teseo 5 STMicro module (DUT #4).

A separate Mosaic X5 receiver, connected via SMA–TNC cabling, functions as the stationary Real-Time Kinematic

```
Root/
|-- Attack Type/
    |-- Rover State (stationary, dynamic)/
        |-- Power Category/
            |-- Bands_X_Y_Z/
                |-- Scenario_ID/
                    |-- scenario.json
                    |-- info.txt
                    |-- mon_rf.csv
                    |-- nav_pvt.csv
                    |-- rinex.csv
```

Fig. 4. Hierarchical folder structure of the open-source GNSS-attack dataset.

Positioning (RTK) base station. All RF connections employ sealed bulkheads and elastomeric mounts to isolate vibration, preserving signal integrity and ensuring time-synchronized logging across all devices.

D. Open-Source Dataset Description

The dataset is released as an open source and is organized in a hierarchical folder structure that reflects the various attack scenarios and their associated properties [22]. These data were collected with a ZED-F9P receiver at a 5 Hz sampling frequency [23]. The folder tree is illustrated in Fig. 4.

Each scenario folder contains the following.

1) *scenario.json:* Metadata including attack type, frequency bands, power settings, rover state, and other scenario-specific parameters.
2) *info.txt:* A summary outlining the key properties of the scenario.
3) *Data Files:*
    1) `mon_rf.csv`*:* Radio Frequency (RF) monitoring data.
    2) `nav_pvt.csv`*:* Navigation Position, Velocity and Time (PVT) data.
    3) `rinex.csv`*:* GNSS observation data in RINEX format.
    4) Additional CSV and UBX files containing raw GNSS measurements.

A Python notebook is provided to facilitate data exploration and analysis. This notebook includes modules to generate various plots and visualizations, such as

1) Plots of SNR metrics for different satellite constellations, in addition to carrier phase, Doppler, and pseudorange values.
2) Analysis of navigation parameters including ground speed, horizontal accuracy, and position Dilution of Precision (DOP).
3) Visualization of the I and Q magnitude and offset measurements.
4) Analysis of Automatic Gain Control (AGC) levels and noise level parameters.
5) 3-D correlation plots for each satellite, showing the relationship between pseudorange and Doppler.

This dataset, together with the analysis notebook, is published as open source to support research into the impact of different attack scenarios on GNSS receivers and to develop and validate detection and mitigation strategies.

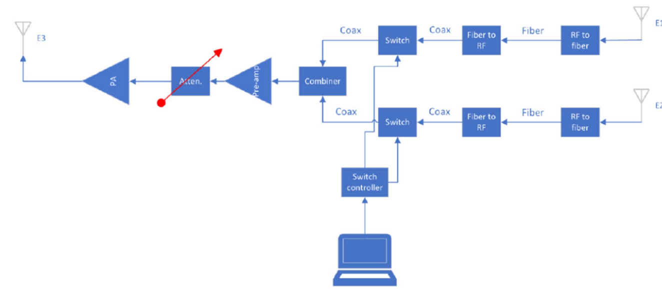


Fig. 5. Schematic diagram of the F1.1 meaconing setup, showing two receiving antennas (E1 and E2) and the directional antenna (E3).

Fig. 6. BladeRF x115 Software-Defined Radio used for the mobile spoofing tests.

E. Dataset Analyze

The dataset contains 24 scenarios, which are divided into four main classes: jamming, spoofing, meaconing and combined jamming and spoofing. For each scenario, we extract a comprehensive set of observation values per satellite: SNR, carrier phase, Doppler, and pseudorange residuals along with receiver metrics such as AGC levels and noise floor estimates. In addition, correlations between pseudorange and Doppler are computed for each tracked satellite to visualise the distortions introduced by each type of attack.

1) Jamming: The following subsections describe the different signatures of jamming signals in three different scenarios: a dynamic rover with a static jammer, a dynamic rover with a dynamic jammer and a static rover with a static jammer.

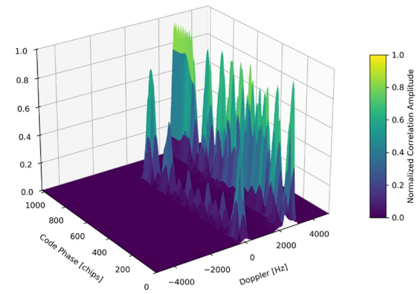
a) Scenario 1. Dynamic rover with static jammer: In this scenario, the SNR exhibits a fluctuating pattern, continuously oscillating up and down for each satellite. Fig. 7 shows the Galileo L1 and L2 SNR values for several satellites during the jamming period. The red shaded areas indicate when jamming was active and show significant SNR fluctuations and periodic signal loss patterns. These characteristics are due to multipath effects in combination with the dynamic motion of the rover relative to a static jammer. Interruptions in Doppler, carrier phase, and pseudorange measurements were also observed. Analysis of the correlation between Doppler and phase code delay shows a high correlation over the entire delay range at some Doppler

Fig. 7. Galileo L1 and L2 SNR values for multiple satellites during the jamming period. The red-shaded areas indicate when jamming was active.

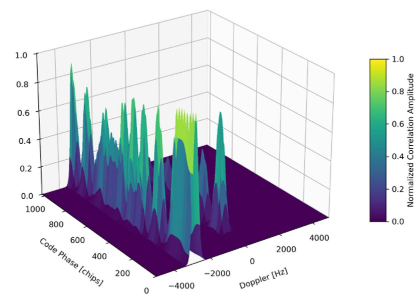
frequencies. This broad correlation is a hallmark of jamming, suggesting that the jammer affects a wide range of delay values (Fig. 8). The AGC drops sharply to a low value at the beginning of the jamming and then fluctuates as the rover is dynamic and thus also has the multipath effect. Upon cessation of jamming, the AGC returns to a higher level, which is consistent with weak satellite signals. The noise is inversely related to the AGC, i.e. it increases when the AGC decreases, reflecting the influence of the jamming signal (Fig. 9). The distinctive feature of this scenario is the continuous SNR fluctuations accompanied by periodic satellite losses, making the swinging SNR pattern and the AGC fluctuations key detection indicators.

b) Scenario 2. Dynamic rover with dynamic jammer (car ahead or behind): When both the rover and the jammer are in motion, such as a jammer in a car moving in front of or behind it, the observed jamming characteristics differ. The SNR still experiences a sudden drop, but with fewer oscillations compared to the scenario with a static jammer. The disruption pattern is more stable. Similar to the previous scenario, interruptions occur in Doppler, carrier phase, and pseudorange measurements. The correlation analysis also shows a high correlation over the entire delay range at specific Doppler frequencies, indicating a broad effect of the jammer. The AGC drops suddenly, but shows fewer oscillations than with a static jammer. The drop is pronounced and stable. The noise levels again inversely mirror the AGC behavior, increasing as the AGC decreases. The more stable interference pattern is a distinctive feature of this scenario, and a stable AGC drop with fewer oscillations serves as an important detection indicator.

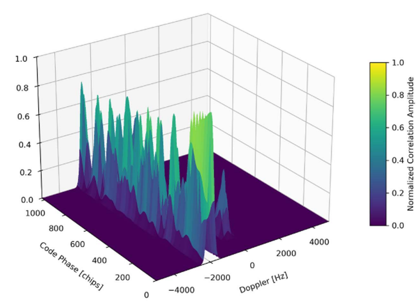
c) Scenario 3. Static rover with static jammer: The impact of jamming is most severe when both the rover and the jammer are stationary. The SNR drops drastically, leading to a complete loss of signal lock. The AGC gradually decreases to a very low value and remains there until the jamming ceases. Recovery of the AGC to normal levels is also a slow process. Similarly to dynamic rover scenarios, the correlation between Doppler and phase delay shows a spread, but it is not as extensive



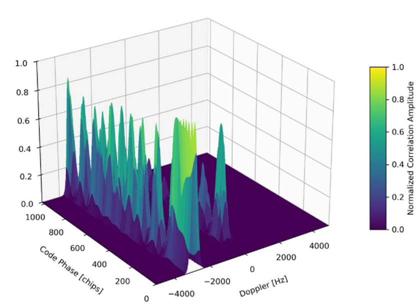
(a)



(b)



(c)



(d)

Fig. 8. Correlation plots for Galileo satellites E03, E06, E09, and E34 over different code phases (chips) and Doppler frequencies (Hz). These plots show the distinctive jamming signature, which is characterized by a high correlation over wide ranges of delay values at certain Doppler frequencies and indicates jamming interference patterns. (a) E03. (b) E06. (c) E09. (d) E34.

TABLE II
COMPARISON OF GNSS SIGNAL BEHAVIOR UNDER DIFFERENT JAMMING SCENARIOS

Metric	Scenario 1: Dynamic Rover, Static Jammer	Scenario 2: Dynamic Rover, Dynamic Jammer	Scenario 3: Static Rover, Static Jammer
SNR Behavior	Fluctuates in a "swinging pattern" with continuous ups and downs	Less oscillation compared to Scenario 1, more stable disruption pattern	Drastic drops leading to complete loss of signal lock
Doppler and other Measurements	Interruptions in Doppler, carrier phase, and pseudorange	Similar interruptions but with more consistent patterns	Complete loss of measurements during intense jamming
AGC Response	Sharp drops with significant fluctuations	Sudden drop but with fewer oscillations	Gradual decrease to very low values with slow recovery
Noise Level	Inversely mirrors AGC, increases during jamming	Similar inverse relationship with AGC	Significant increase during jamming, mirrors AGC drop
Correlation Pattern	High correlation across entire delay range at specific Doppler frequencies	Similar broad correlation but more consistent	Spread correlation but less extensive than with dynamic jammer

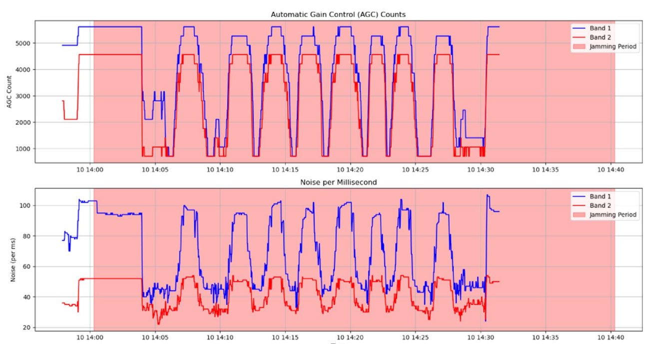


Fig. 9. AGC levels and noise per millisecond measurements during the jamming period.

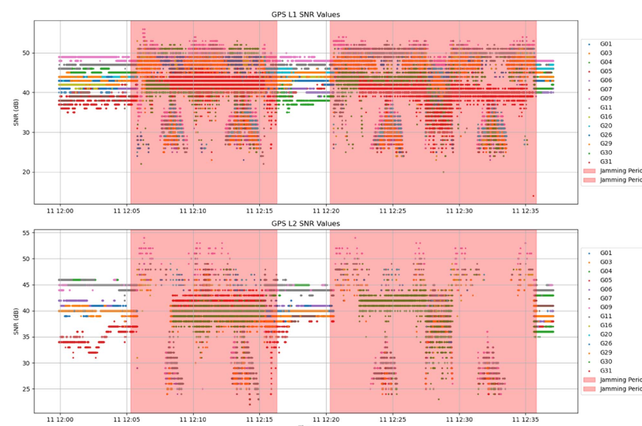


Fig. 10. GPS L1 and L2 SNR values across multiple satellites during spoofing periods (highlighted in red).

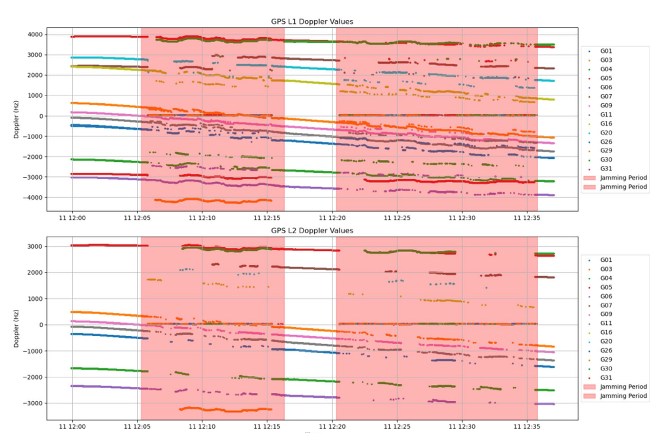


Fig. 11. GPS L1 and L2 Doppler values across multiple satellites during spoofing periods (highlighted in red).

as with a dynamic jammer. The impact in this case is more stable and persistent.

Analyzing these three distinct jamming scenarios shows that each of them produces unique and identifiable patterns in critical GNSS signal metrics. By observing the behavior of SNR, Doppler measurements, AGC, noise levels, and correlation patterns, it is not only possible to detect the presence of jamming signals, but also to potentially discern the nature of the jamming scenario, including the relative motion between the receiver and the jammer. The summary of these distinctive patterns is provided in Table II. This understanding is crucial for developing effective anti-jamming techniques and mitigation strategies.

2) Spoofing: The impact of spoofing on GNSS signals varies significantly depending on the relative motion of the rover and the spoofer. The following subsections detail the distinct signatures observed in two different scenarios: A dynamic rover with a stationary spoofer and a static rover with a stationary spoofer.

a) Spoofing Scenario 1. dynamic rover—stationary spoofer: The SNR exhibits a sudden transition, either increasing or decreasing depending on the satellite being spoofed. This transition often involves fluctuations, but the dominant feature is a sharp swing in SNR values at the beginning of spoofing. Fig. 10 clearly shows these characteristic SNR fluctuations during the spoofing event. There is a sudden transition in the Doppler measurements, accompanied by notable fluctuations and discontinuities (Fig. 11). The AGC drops sharply to a low value when the spoofer starts and then fluctuates. After the interference stops, the AGC returns to a higher value, which is typical for weak satellite signals. Noise inversely mirrors the behavior of the AGC: while the AGC decreases, the noise increases, reflecting the influence of jamming (Fig. 12). A key pattern emerges during spoofing, revealing two dominant correlation peaks: one with a

TABLE III
COMPARISON OF GNSS SIGNAL BEHAVIOR UNDER DIFFERENT SPOOFING SCENARIOS

Metric	Scenario 1: Dynamic Rover, Stationary Spoofer	Scenario 2: Stationary Rover, Stationary Spoofer
SNR Behavior	Sudden transition (increase or decrease) with fluctuations and sharp swings at the beginning of spoofing	Sudden transitions accompanied by small oscillations or ripples, indicating signal interference and inconsistent reception
Doppler and other Measurements	Sudden transition with notable fluctuations and discontinuities	Even sharper transitions than in dynamic scenarios; abrupt fluctuations, high discontinuities, and occasional loss of measurements from some satellites
AGC Response	Sharp drops to a low value when spoofing starts, subsequently fluctuates, returns to higher level after spoofing ends	Sharp decrease at the start of spoofing, smooth and free of fluctuations, indicating a cleaner but more powerful spoofing input
Noise Level	Inversely mirrors AGC behavior: As AGC drops, noise increases, reflecting the influence of jamming	Inversely mirrors AGC behavior, increasing whenever AGC falls
Correlation Pattern	Two dominant correlation peaks: One with very high correlation to the spoofed signal, another with lower correlation to the authentic signal	More than two correlation peaks appear, with spoofing peak dominating over a wider phase delay range, making mitigation much less reliable

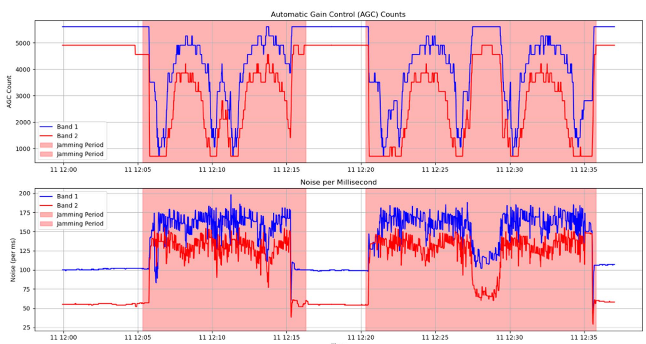


Fig. 12. AGC levels, and noise per millisecond measurements during spoofing periods (highlighted in red).

very high correlation to the spoofed signal and another with a lower correlation that typically corresponds to the authentic signal. These two peaks indicate competing signal sources (fake and authentic satellite signals) with time-dependent variations in signal strength. Fig. 13 shows the correlation patterns for the four satellites under spoofing conditions. The overall effect is a fluctuating position control. The distinctive feature of this scenario is the competition between signal sources with time-dependent strength fluctuations.

b) Spoofing Scenario 2. Stationary rover—stationary spoofer: The SNR exhibits sudden transitions accompanied by small oscillations or ripples. These drops indicate signal interference and inconsistent reception. Transitions in Doppler and other measurements are even sharper than in dynamic scenarios: they fluctuate abruptly, show high discontinuities, and occasionally lose measurements from some satellites. AGC values drop sharply at the start of spoofing. Unlike the dynamic case, this decrease is smooth and free of fluctuations, indicating a cleaner but more powerful spoofing input. Noise mirrors the behavior of AGC in an inverse way, increasing whenever AGC falls. In the stationary scenario, more than two correlation peaks appear and the spoofing peak dominates over a wider phase delay range, making detection much less reliable. Overall, these effects are more severe and stable, giving attackers greater control over positioning without needing to compensate for rover movement, yielding a more pronounced spoofing signature.

Analyzing these two different spoofing scenarios shows that each of them generates unique and identifiable patterns in critical GNSS signal metrics. By observing the behavior of SNR, Doppler measurements, AGC, noise levels and correlation patterns, it is not only possible to detect the presence of spoofing, but also to potentially discern the nature of the spoofing scenario, including the relative motion between the rover and the spoofer. The summary of these characteristic patterns can be found in Table III. This understanding is crucial for the development of effective antispoofing techniques and mitigation strategies.

3) Meaconing: In this section, GNSS meaconing is analyzed by quantifying its impact on the receiver observables SNR, Doppler/code measurements, AGC, noise, and delay. Similar to jamming and spoofing, the kinematic relationship between receiver and meaconer together with the rebroadcast power level provides characteristic signatures that allow the scenarios to be distinguished.

In all measurement tests, the retransmitted signal leads to an abrupt degradation of the SNR, accompanied by intermittent loss of satellite tracking and sporadic gaps in the Doppler, carrier phase and pseudorange data. When the rover remains stationary, these dropouts are sharper and occur more persistently. The AGC of the receiver drops rapidly at the beginning of the meaconer. In contrast to wide-band jamming, the AGC shows only small residual oscillations in both the dynamic and static cases, reflecting the quasi-continuous, narrowband nature of the rebroadcast. Delay–Doppler correlation maps show several secondary peaks that are broader than the discrete spoofing lobes but far less extensive than the flat high-density plateaus produced by jamming, confirming the intermediate spectral footprint of meaconing.

V. MITIGATION STRATEGIES AND ALTERNATIVES

Much of the research is dedicated to the development of resilient GNSS solutions. To counter GNSS-based applications from jamming and spoofing attacks, security measures such as location data authentication, cryptographic techniques for signal authentication, continuous monitoring of computer networks, anomaly detection, sensor fusion (inertial navigation systems,

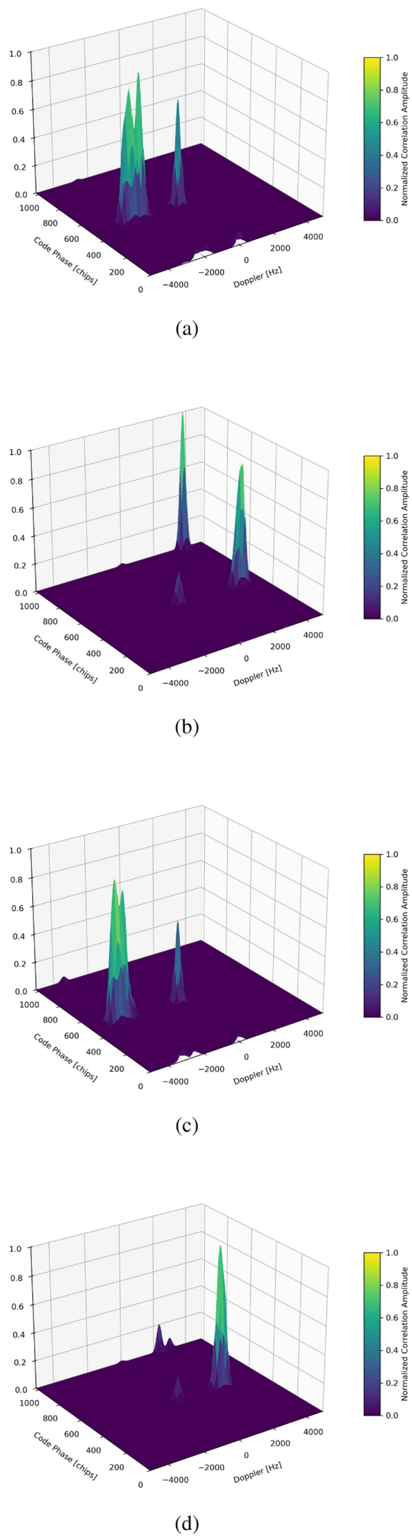


Fig. 13. Correlation plots for GPS satellites G04, G05, G06, and G07 during spoofing events in the Dynamic Rover—Stationary Spoofer scenario. Each 3-D plot shows the correlation amplitude across code phase (chips) and Doppler frequency (Hz). The dual peaks visible in each plot represent the competing signal sources: One peak with higher correlation (spoofed signal) and another with lower correlation (authentic satellite signal). (a) G04. (b) G05. (c) G06. (d) G07.

cameras), game theory-based strategies for strategic maneuvers, and machine learning algorithms for spoofing pattern detection are adopted. The strategies are developed at both technological and political levels. This section does not provide an exhaustive analysis of all existing approaches, but outlines and illustrates the main categories of approaches.

A. Technological Countermeasures

Anti-jamming antennas and receivers are designed to filter out or ignore jamming signals. [24] proposes a Jamming Aware Artificial Potential Field (JA-APF) method to mitigate GNSS jamming attacks on unmanned surface vessels by improving path planning through detection, localization, and adaptive re-planning. JA-APF detects jamming by monitoring anomalies in signal strength and error rates and localizes the source of interference. The path is then dynamically recalculated to navigate away from the threat. Both attractive and repulsive potential fields are used to guide the ship safely to its destination. Performance evaluations were only carried out in the simulation.

With an average of 3.2 million smartphones sold every day [25] and around 30% of smartphone apps requiring access to location data [26], 960 000 people run the risk of using location-based services (LBS) with false or even hacked information every day. In this context, algorithms are developed to protect users. GNSSAlarm [27] is a set of algorithms that use the existing resources of the Android operating system to provide real-time alerts for jamming and spoofing. Four methods detect attacks by monitoring various metrics, including network location metrics, AGC and C/NO metrics, external aiding information, and pseudorange residual metrics. All of these methods depends on the analyze of measurement that we have seen in Section IV. The first method checks the discrepancy between GNSS-based and network-based (Wi-Fi and cell towers) location estimates. The second method monitors whether the location is overridden by a mock provider application using the Android flag *isFromMockProvider()*. A third method checks synchronization discrepancies between GNSS and system time. The fourth method uses AGC and C/NO values to distinguish between jamming and spoofing. AGC decreases when jamming occurs, while C/NO can indicate spoofing. Experimental jamming detection tests in the GNSS L1 band were successfully performed with a Samsung 20+. An AGC dB loss threshold was used, which must be adapted to the specific phone type. A spoofing test with a change of location was correctly detected with the first method.

A further countermeasure, demonstrated on scale during the war in Ukraine, is the move to fiber optic-tethered drones. These systems represent a robust solution against GNSS jamming and spoofing attacks by maintaining continuous communication with ground control stations through physical fiber optic cables rather than radio frequencies. Wired guidance makes the link between operators and drones immune to jamming and allows for much faster and better quality updates from the drone, even from locations where radio contact would be poor, without revealing operator's or drone's location by radio signals [28].

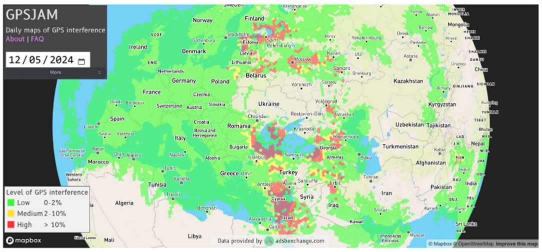


Fig. 14. GPSJam Map of areas where a significant percentage of aircraft report low GNSS navigation accuracy on May 12, 2024.

These systems navigate through real-time command relay from ground-based operators who can utilize optical navigation system, including surveyed landmarks, to estimate positioning and transmit steering commands via the fiber link [29]. The tethered approach provides complete immunity to electronic warfare tactics targeting GNSS and wireless systems, though operational range is limited to the fiber cable length, typically 2–25 km [30].

Some initiatives aim at continuously monitoring geographical areas affected by GNSS interference to share this information with the international community and issue warnings. The GPSJam website [31] uses digital radio messages (ADS-B) broadcast by aircraft containing the accuracy of the on-board GNSS navigation system to create a map showing the zones of potential navigation system interference observed in the last 24 hours (Fig. 14). The messages are extracted from the ADS-B Exchange Platform. Founded in 2016 by Dan Streufert, this platform is a crowdsourced flight data exchange platform supported by a worldwide network of volunteers who operate signal receivers and collect data from nearby aircraft [32]. GPSJam map green hexagons indicate areas where more than 98% of the aircraft that flew over the area reported good GNSS positioning accuracy. Yellow hexagons correspond to areas where 2% to 10% of aircraft reported poor GNSS positioning accuracy. Red hexagons indicate areas where more than 10% of aircraft reported poor GNSS positioning accuracy.

B. Policy and Regulatory Measures

The effective management and mitigation of GNSS interference risks require robust policy and regulatory measures. To this end, international cooperation is essential to regulate the manufacture and sale of jamming equipment and to limit the availability of devices that can interfere with critical navigation systems. In addition, policies that encourage the development and widespread adoption of safer navigation technologies are crucial. These policies not only improve the resilience of GNSS systems against threats but also encourage innovation in the technology sector.

In the US, the Federal Communications Commission (FCC) has established regulations that restrict the use of non-GPS GNSS signals without prior FCC approval. This regulation specifically allows the use of certain signals from the European Galileo system, as Galileo is the only system that has applied for and received authorization. Consequently, any operational use of other GNSS signals, such as the Chinese BeiDou, the Russian GLONASS, unapproved signals from Galileo or the Japanese QZSS, is technically unlawful without the approval of the FCC. This situation poses a major challenge, as the integration of multiple GNSS systems is widely accepted and can improve the accuracy and reliability of positioning. Device manufacturers want to produce universally compatible devices without having to adapt the hardware and software for specific regional regulations. In addition, there are security concerns related to the use of GLONASS and BeiDou systems that make their introduction and approval in the US difficult. This regulatory environment not only has an impact on the technological landscape, but also on strategic decisions in the telecommunications and navigation industry.

C. Innovation in Navigation Technologies

As for the vulnerabilities of GNSS, it is important to recognize that the signals of all satellite constellations can potentially be compromised, so there is no inherent security advantage to using one system over another. For example, if hostile countries attack a country that uses GNSS signals, they may prefer to use fake GNSS signals generated by the enemy satellite constellation that could be mistaken for real enemy signals rather than using their own GNSS signals. In this context, combining multiple GNSS systems and confronting them can also be a strength.

Another approach to address the vulnerabilities of using foreign GNSS signals for safety-critical applications is to turn them into technical risks [33]. A technical framework is established to define specific technical failures or anomalies that occur with interference and analyze them using standard risk assessment techniques such as the likelihood and consequences (5x5 risk matrix). The potential impact of the identified GNSS signal anomalies on military and civilian applications is analyzed using computational simulations and radio frequency (RF) signal constellation simulators. This approach helps to quantify the impact on the position, velocity, and time (PVT) solutions offered by GNSS receivers, including their built-in anti-spoofing (AS) and other signal integrity mechanisms.

Given this universal vulnerability, cybersecurity focuses less on the source of the signals and more on the receivers themselves. Indeed, ensuring the integrity and security of GNSS receivers is becoming increasingly important and requires stringent security measures throughout the design and manufacturing process to defend against malicious attacks that exploit vulnerabilities in GNSS signals. Some encrypted GNSS signals exist. The GPS P(Y) code is for military purposes. For Galileo, the Public Regulated Service (PRS) is restricted to government-authorised users requiring a high level of service continuity and the Open Service Navigation Message Authentication (OSNMA) is a data authentication function based on the existing lightweight standard broadcast authentication protocol named TESLA (Timed-Efficient Stream Loss-Tolerant Authentication), which is freely available to all [34], [35]. In addition, there is the High Accuracy Service (HAS), which provides highly accurate corrections via the Galileo E6-B data component and terrestrial channels to achieve a positioning accuracy of two decimeters under normal conditions.

Machine learning and deep learning detect and mitigate GNSS jamming/spoofing by learning complex, nonlinear relationships in receiver observables; supervised classifiers over metrics (e.g., C/N_0, AGC, pseudorange residuals) provide real-time interference flags [36]. Deep models operate directly on correlation and delay–Doppler representations to discriminate authentic from counterfeit replicas [37], [38]. Privacy-preserving self-supervised and federated schemes learn spoofing deviations without centralized labels, enabling edge deployment [39]. In practice, these detectors feed integrity monitoring systems, trigger non-GNSS systems (e.g., INS/UWB/vision), and, where available, drive antenna-array beamforming/null-steering [40], [41], [42].

Finally, the combination of GNSS-based and non-GNSS solutions, many of which have been developed by the IPIN research community, is a very interesting strategy. The best-known example is the use of multi-sensor navigation systems that combine GNSS with inertial navigation systems. They are often used for autonomous car navigation and indoor robotic applications.

D. IPIN Research: A Major Source of Alternative Positioning

Research within the IPIN community has historically focused on developing robust indoor positioning systems, tailored for environments where GNSS fails or is unavailable. However, as the vulnerability of GNSS in outdoor and mission-critical environments grows, the technologies and methods pioneered by IPIN are now being recognized for their broader potential to support resilient, seamless indoor-outdoor positioning and navigation (PNT) systems. The following review, based on the best paper contributions from the past five IPIN conferences, highlights innovations with strong relevance for multienvironment deployment and GNSS-resilient PNT.

Several approaches use radio signals, specifically designed to address propagation challenges in urban and indoor environments, employing high-frequency pulses across broad bands or even radar signals. By using the Time Difference of Arrival of ultrawideband (UWB) signals, [43] improved positioning accuracy in large areas while emphasizing scalability and robustness. The authors in [44] incorporated cascaded wireless clock synchronization to overcome timing challenges in multizone environments and [45] further enhanced this approach by combining UWB and radar CIR features to improve drone localization through multimodal sensing. The authors in [46] shifted the focus to easy deployment and scalability of UWB systems. The plug-and-play model enabled rapid deployment without complex preconfiguration, offering a pathway toward portable and adaptable positioning solutions suitable for dynamic outdoor or semistructured environments.

Other approaches exploit more natural signals in the urban environment, such as light or sound. [47] explores Visible Light Communication for high-precision tasks such as landing drones with the modulation of light sources. This is particularly relevant for GNSS-denied outdoor scenarios such as dense urban canyons or seamless indoor-to-outdoor transitions where radio communication is degraded. [48] leverages sparse acoustic arrays to estimate the Direction of Arrival (DoA) of sounds, which is applied for indoor localization and navigation by detecting sound sources or echoes within a building. Practical tests of array geometries and DoA algorithms are carried out with real acoustic data. The authors in [49] proposed a joint calibration method for microphone arrays and sound source localization using information-aware modeling, enhancing robustness in microphone-based systems with minimal prior setup while [50] built on acoustic fingerprinting and introduced multipath-Time-of-Flight (ToF) profiles with CNN regression, significantly improving accuracy over traditional PSD-based methods, opening opportunities for sound-based positioning in complex transitional environments.

Machine learning and deep learning techniques are being used more and more to increase robustness. The authors in [51] used advanced machine learning models such as transformers to estimate multipath delays that are reflected from different surfaces and distort the positioning. This approach mitigates the classical difficulty of multipath effects in complex environments. Likewise, [52] enhanced indoor radio maps using machine learning, enabling dynamic adaptation to environmental changes, which is a cornerstone for long-term deployment of resilient positioning infrastructures. In the RF domain, [53] applied a time-invariant convolutional neural network to CSI-based human localization, achieving high accuracy even in noisy conditions. Finally, [54] addressed magnetometer calibration in SLAM by eliminating the need for manual device rotation, facilitating seamless integration in dynamic systems.

These award-winning contributions from the IPIN community demonstrate the rapid development of positioning technologies beyond static indoor use cases. Their scalability, adaptability and potential to be deployed with low infrastructure overheads make them powerful enablers for seamless indoor-outdoor navigation. As the resilience of GNSS-based systems becomes a national and industrial priority, these IPIN-derived methods provide concrete, field-proven building blocks for robust and secure PNT solutions in GNSS-challenged environments.

VI. Conclusion

This study provides documentation of the growing threat to GNSS, with examples drawn from current events and tests carried out in Norway in 2024. It also discusses the importance of the IPIN research outcomes in answering the critical need for resilient alternative navigation solutions. Our analysis reveals a dramatic increase in GNSS attacks, with the number of interferences increasing sevenfold from 50 perturbations per week in 2023 to over 350 per week in 2024, affecting 46 000 flights over the Baltic Sea region alone. The military implications are severe, with GPS-guided munitions experiencing accuracy degradation from 50% to less than 10% effectiveness due to systematic jamming.

This sharp increase in GNSS attacks has motivated us to collect real-world data. We collected 24 carefully staged attack scenarios over a period of 819 minutes, with jamming signals dominating the dataset with 54% of the total attack time. Pure spoofing accounted for 27%, spoofing + jamming for 10% and meaconing for 9%. Overall, the attacks occurred predominantly in dynamic scenarios: 69% dynamic compared to 31% stationary. Furthermore, we analyzed this new experimental data and defined a unique fingerprint for each attack. The full dataset and analysis code are published as open source.

The IPIN community's research, initially focussed on indoor environments, developed novel approaches that can be considered central to countering outdoor vulnerabilities. Analyzing the top contributions over the last five editions of the IPIN conference shows that advanced techniques such as sensor fusion and signal processing are improving the resilience of navigation systems in areas such as smart cities and autonomous vehicles. By linking indoor and outdoor challenges, IPIN's work is seen as essential to overcoming GNSS vulnerabilities and ensuring global navigation security.

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